

Algebra

Volume 2

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mapping $i: X \rightarrow F_X$. We also recall that an object defined by a universal mapping property is determined up to a unique isomorphism (Vol. 1, p. 97).

References to the bibliography are by the name of the author in small capitals, with the date to distinguish works by the same author. As in Vol. 1 we use iff as an abbreviation for 'if and only if' and ■ indicates the end (or absence) of a proof.

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Natural numbers, cardinal numbers, ordinal numbers

In Volume 1 we assumed the natural numbers as known and freely used their properties without proof. If our development is to have any claim to completeness we shall need either to provide an axiomatic foundation for the natural numbers, or to deduce their properties from some other domain such as set theory. The latter alternative would of course make it necessary to provide an axiomatic foundation of set theory and this would involve us far deeper in foundational questions than is appropriate here. Such a study would occupy a whole volume by itself, and would not greatly help us in our understanding of algebra. We shall therefore confine ourselves to a derivation of the properties of natural numbers from the Peano axioms and a brief discussion of their significance, as well as their relevance to algebra. The remainder of the chapter describes the methods of dealing with infinite sets.

1.1 The Peano axioms

We begin by writing down a system of axioms for the natural numbers; this is not so much to give a rigorous foundation, as to make explicit the properties of numbers we are using. The notions of set theory will be used freely, in the intuitive form introduced earlier. We shall also use individual numbers (e.g. to label the axioms) without hesitation; clearly no axioms are needed for the numbers up to 12, or for that matter up to 10^{100} . The purpose of the axioms is to allow us to deal with the set of *all* numbers.

The axioms, as stated essentially by Peano in 1889, are:

- N. 1 *1 is a natural number.*
- N. 2 *Every natural number n has a unique successor n' , which is again a natural number.*
- N. 3 *1 is not the successor of any number.*
- N. 4 *Distinct numbers have distinct successors: $m \neq n \Rightarrow m' \neq n'$.*
- N. 5 *(Principle of induction): A set of natural numbers containing 1 and with each number its successor contains all natural numbers.*

2 Natural numbers, cardinal numbers, ordinal numbers

The set of all natural numbers will be denoted by $\mathbf{N}$; by $\mathbf{N}$, 2 we may think of 'as a unary operation (i.e. a function of one argument): $a \mapsto a'$ '. Let us call a set with a unary operation an *induction algebra* for the moment. This is an algebraic system, like a monoid or a group, and as in that case we can define the notions of homomorphism, subalgebra and generating set. Thus a subalgebra of an induction algebra is a subset which admits the unary operation; e.g. $\mathbf{N}$, 2 and $\mathbf{N}$, 5 may be expressed by saying that $\mathbf{N}$ is an induction algebra generated by the single element 1.

Our first aim is to prove a certain universal mapping property of $\mathbf{N}$, which forms the basis of the process of definition by recursion. The following lemma will be needed in the proof of Th. 2:

LEMMA 1 *Let A be an induction algebra and $b \in A$, then the subalgebra B generated by b consists of b and successors of elements of B .*

For let B_1 be the set consisting of b and successors of elements of B . Clearly $B_1 \subseteq B$, and B_1 contains b and the successor of any element in it, hence B_1 is a subalgebra. Since B is the least subalgebra containing b , it follows that $B_1 = B$. ■

For example, if we take $A = \mathbf{N}$, $b = 1$, and remember $\mathbf{N}$, 5, the lemma tells us that every number different from 1 is the successor of a number.

THEOREM 2 *Let A be an induction algebra and a an element of A ; then there is a unique homomorphism $h: \mathbf{N} \rightarrow A$ such that $h(1) = a$.*

In detail the assertion states that A is a set with a unary operation $x \mapsto x'$, and given $a \in A$, there is a unique mapping $x \mapsto h(x)$ from $\mathbf{N}$ to A such that

$$h(1) = a, \quad h(x') = h(x)'.$$

There can be at most one such mapping, because a homomorphism is determined by its values on a generating set (cf. Vol. 1, Lemma 3 of 6.4). To prove its existence, we form the direct product $\mathbf{N} \times A$. This is again an induction algebra, if we define the operation componentwise: $(x, y)' = (x', y')$. Let H be the subalgebra of $\mathbf{N} \times A$ generated by $(1, a)$; further denote by p, q the projection mappings of $\mathbf{N} \times A$ on the factors $\mathbf{N}$ and A respectively, and by p', q' their restrictions to H .

The image of H under p' is a subalgebra of $\mathbf{N}$ containing 1, and hence is $\mathbf{N}$ itself, by $\mathbf{N}$, 5. Thus for each $x \in \mathbf{N}$,

$$\text{there exists } y \in A \text{ such that } (x, y) \in H. \quad (1)$$

We claim that for each $x \in \mathbf{N}$ there is just one such y . For by Lemma 1, H consists of $(1, a)$ together with successors of elements of H . Any successor is of the form (x', y') ; here the first component is different from 1, by $\mathbf{N}$, 3, and so the y determined in (1) by $x = 1$ is unique. Let $\mathbf{N}_0$ be the subset of all $x \in \mathbf{N}$ occurring as first component in just one element of H , i.e. for which the $y \in A$

1.1

The Peano axioms

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determined in (1) is unique. As we have seen, $1 \in \mathbf{N}_0$; if we can show that $\mathbf{N}_0$ contains with each element its successor, it will follow from $\mathbf{N}$, 5 that $\mathbf{N}_0 = \mathbf{N}$.

Let $x \in \mathbf{N}_0$ and suppose that $y_1, y_2 \in A$ are such that $(x', y_i) \in H$ ($i = 1, 2$); we have to show that $y_1 = y_2$. Since $x' \neq 1$, (x', y_i) is a successor in H , say $(x', y_i) = (u_i, v_i)' = (u_i', v_i')$, where $(u_i, v_i) \in H$. Equating first components, we find $x' = u_1' = u_2'$, and by $\mathbf{N}$, 4, $x = u_1 = u_2$, i.e. $(x, v_i) \in H$ for $i = 1, 2$. But $x \in \mathbf{N}_0$ and this means that $v_1 = v_2$; hence $v_1' = v_2'$ and so $y_1 = y_2$ as we had to show. Therefore $x' \in \mathbf{N}_0$ and by $\mathbf{N}$, 5 we conclude that $\mathbf{N}_0 = \mathbf{N}$.

Thus we have shown that for each $x \in \mathbf{N}$ there is a unique $y \in A$ such that $(x, y) \in H$. Writing $h(x)$ for y , we find that $h(1) = a$, $h(x') = h(x)'$, which is what we had to prove. ■

The result may be expressed by saying that $\mathbf{N}$ has a universal mapping property for induction algebras (cf. Vol. 1, 4.9 and p. viii of this vol.); it shows in particular that the properties of $\mathbf{N}$ determine it as an induction algebra, up to isomorphism. By analogy with modules (Vol. 1, Ch. 10) or groups (Vol. 1, Ch. 9), $\mathbf{N}$ may be described as the *free induction algebra* on 1 as a *free generator*.

Functions on $\mathbf{N}$ are frequently defined recursively, for example the sum of the squares of the first n natural numbers may be defined as the function $g: \mathbf{N} \rightarrow \mathbf{N}$ such that

$$g(1) = 1, \quad g(n+1) = g(n) + (n+1)^2.$$

It may appear intuitively obvious that this defines a function, but this needs to be proved. The basic reason is that $\mathbf{N}$ is a free induction algebra on 1 as free generator, as we shall now see. The statement in Th. 3 is a little more general, but the proof begins with the case exemplified above.

THEOREM 3 *Given $a \in \mathbf{N}$ and any function f from $\mathbf{N}^2$ to $\mathbf{N}$, there exists a unique function φ from $\mathbf{N}$ to itself, satisfying the equations:*

$$\varphi(1) = a, \quad \varphi(n') = f(n, \varphi(n)). \quad (2)$$

Proof. Suppose first that f is independent of its first argument, then we have to find $\varphi: \mathbf{N} \rightarrow \mathbf{N}$ to satisfy

$$\varphi(1) = a, \quad \varphi(n') = f(\varphi(n)). \quad (3)$$

In this case the result follows immediately from Th. 2, taking A there to be the set $\mathbf{N}$ with f as its successor function.

In the general case we take $A = \mathbf{N}^2$ with successor function

$$(x, y) \mapsto (x', f(x, y)),$$

and apply Th. 2 with the element $(1, a)$ in place of a . We obtain a mapping $\varphi: \mathbf{N} \rightarrow \mathbf{N}^2$; if its projections on the factors are φ_1 and φ_2 , then $\varphi_1(1) =$

1, $\varphi_1(n) = n'$, hence $\varphi_1(x) = x$ for all $x \in \mathbf{N}$, and now

$$\varphi_2(1) = a, \quad \varphi_2(n) = f(n, \varphi_2(n')).$$

Thus φ_2 is the required function. ■

This result emphasizes the difference between 'proof by induction' and 'definition by recursion'. Whereas the former is embodied in N. 5, the latter relies also on N. 3, 4, and a further argument (such as that leading to Th. 3) is needed to prove it. From the algebraist's point of view the situation can be summed up by saying that a *proof* by induction depends on the fact that $\mathbf{N}$, as induction algebra, is generated by 1, whereas the method of definition by recursion depends on the fact that $\mathbf{N}$ is generated *freely* by 1.

As an application let us see how Th. 3 may be used to define addition and multiplication on $\mathbf{N}$. Given $a \in \mathbf{N}$, there is a mapping $\alpha_a: \mathbf{N} \rightarrow \mathbf{N}$ such that

$$\alpha_a(1) = a', \quad \alpha_a(x') = \alpha_a(x)'$$

If we write $a + x$ in place of $\alpha_a(x)$, these equations take on a more familiar form:

$$a + 1 = a', \quad a + x' = (a + x)'. \quad (4)$$

From this definition it is easy to prove the associative and commutative laws of addition:

$$(a + b) + c = a + (b + c), \quad (5)$$

$$a + b = b + a. \quad (6)$$

We shall prove (5) as an example and leave (6) to the reader. For $c = 1$, (5) reduces to $(a + b) + 1 = a + (b + 1)$, i.e. $(a + b)' = a + b'$, which is true by the definition (4). If we assume that (5) holds for $c = n$, then $(a + b) + n' = [(a + b) + n]' = [a + (b + n)]' = a + (b + n)'$, by (4) and the case $c = n$. Hence (5) holds for $c = n'$, and so by induction (N. 5) it holds for all n .

Similarly we can define multiplication by constructing for each $a \in \mathbf{N}$, a mapping $\mu_a: \mathbf{N} \rightarrow \mathbf{N}$ such that $\mu_a(1) = a$, $\mu_a(x') = \mu_a(x) + a$. The existence and uniqueness again follows from Th. 3, and we write as usual $\mu_a(x) = ax$, so that the definition takes the form

$$a1 = a, \quad a(x + 1) = ax + a. \quad (7)$$

The associative and commutative laws can again be proved without difficulty, as well as the distributive laws. If we adjoin a new element, denoted by 0, to $\mathbf{N}$, to satisfy $a + 0 = a$, $a0 = 0$, we have a monoid under addition, and the procedure of Vol. 1, 6.2 enables us to embed it in a group, which is the additive group of integers $\mathbf{Z}$. All that is needed is a proof of the cancellation law: $a + c = b + c$ implies $a = b$; this presents no difficulty and may be left to the reader. Now the multiplication on $\mathbf{N}$ can easily be extended to $\mathbf{Z}$ and the distributive law verified; thus we have obtained the ring structure of $\mathbf{Z}$.

We define an ordering on $\mathbf{N}$, the 'natural' ordering, by putting $m \leq n$ whenever n occurs in the series of successors starting from m : $m, m', m'', \dots$. This may also be expressed by saying that n belongs to the induction subalgebra of $\mathbf{N}$ generated by m . It is not hard to verify that the relation thus defined on $\mathbf{N}$ is a total ordering which is preserved by addition and multiplication:

$$a_1 \leq a_2 \text{ and } b_1 \leq b_2 \text{ imply } a_1 + b_1 \leq a_2 + b_2 \\ \text{and } a_1 b_1 \leq a_2 b_2 \quad (a_i, b_i \in \mathbf{N})$$

It is not difficult to extend the ordering to $\mathbf{Z}$; it is still preserved under addition, but only under multiplication by a positive number.

In algebra the additive and multiplicative structure of $\mathbf{N}$ is naturally of major importance, but the primary purpose of the natural numbers is to count. In essence counting serves to compare the 'numerousness' or 'number-of-elements' of two sets. When Man Friday wanted to tell Robinson Crusoe that he had seen a boat with 17 men in it, he did this by exhibiting another 17-element set, and he was able to do this without being able to count up to 17. Even for a fully numerate person it may be easier to compare two sets rather than to count each; e.g. in a full lecture room a brief glance may suffice to convince us that there are as many people as seats. This suggests that it may be easier to determine when two sets have the same 'number of elements' than finding that number. Let us call two sets *equipotent* if there is a bijection between them. This defines a relation between sets, which on any given collection of sets is an equivalence,[†] so in order to compare two sets, we may compare each to a set of natural numbers. A set S is said to be *finite*, of *cardinal* n , if S is equipotent to the set of natural numbers from 1 to n . It is clear that two finite sets are equipotent if they have the same cardinal, and this may be regarded as the basis of counting. But we need to be assured that sets of different finite cardinals are not equipotent. This may appear intuitively obvious, but it requires proof, and from our axioms such a proof is not hard to find. We first prove a slightly more general result:

THEOREM 4 *Let m, n be natural numbers, then there exists an injective mapping from $\{1, 2, \dots, m\}$ to $\{1, 2, \dots, n\}$ if and only if $m \leq n$.*

Proof. If $m \leq n$, then $x \mapsto x$ provides the required injection. To prove the converse, let us abbreviate $\{1, 2, \dots, v\}$ as $[v]$, for any $v \in \mathbf{N}$. Given an injective mapping $f: [m] \rightarrow [n]$, we must show that $m \leq n$. For $m = 1$ there is nothing to prove, so we assume that $m > 1$ and use induction on m . Since $m > 1$, it is the successor of a number, which we may write as $m - 1$. Now we

[†] We avoid speaking about the collection of all sets, as that would bring us dangerously close to the paradoxes mentioned in Vol. 1, Ch. 1. In any case we shall have no need to do so.

use f to define a mapping $g: [m-1] \rightarrow [n]$ by the rule

$$g(v) = \begin{cases} f(v) & \text{if } f(v) \neq n, \\ f(m) & \text{if } f(v) = n. \end{cases}$$

To check that this is a well-defined mapping we note that there is at most one number v such that $f(v) = n$. Denote this number by v_0 ; if $v_0 = m$, then g is just f restricted to $[m-1]$. Otherwise g differs from f only at v_0 and there it has the value $f(m)$, which it assumes nowhere else (for the domain of g does not include m). Moreover, g does not assume the value n , and it is clearly injective. It follows that $n > 1$ and that g is an injective mapping from $[m-1]$ to $[n-1]$; by the induction hypothesis, $m-1 \leq n-1$; taking successors on each side, we obtain $m \leq n$. ■

If there is a bijection between $[m]$ and $[n]$, we conclude that $m \leq n$ and $n \leq m$, and hence $m = n$. Thus for any finite set, the natural number which indicates its cardinal is uniquely determined.

In contrapositive form Th. 4. shows that if $m > n$, then there can be no injective mapping from $[m]$ to $[n]$. A more illuminating way of expressing this observation is Dirichlet's celebrated

BOX PRINCIPLE (Schubfachprinzip). *If $n+1$ objects are distributed over n boxes, then some box must contain more than one of the objects.*

Although intuitively obvious, this principle is of great use in number theory and elsewhere.

Exercises

- (1) Prove the commutative law of addition on $\mathbf{N}$.
- (2) Prove the associative and commutative laws of multiplication, the distributive law and the cancellation law.
- (3) Give a direct proof that every induction algebra generated by a single element 1 satisfies either $\mathbf{N}$, 3 or $\mathbf{N}_4$.
- (4) For any set S , denote by $\{S\}$ the set whose single member is S (as usual). Let M be the induction algebra generated by $\emptyset$ in this way, by the operation $S' = S \cup \{S\}$. Show that M satisfies $\mathbf{N}_1$ –5 with $\emptyset$ in place of 1.
- (5) Let A be any induction algebra. Show that the union of any two subalgebras is again a subalgebra.
- (6) Let f be a function on the non-negative integers, defined by $f(0) = 0$, $f(x') = x$. Describe the function $g(x, y)$ defined by $g(x, 0) = x$, $g(x, y') = f(g(x, y))$.
- (7) If p_n is the number of equivalence relations on a set of n elements, obtain the following recursion formula for p_n (where $p_0 = 1$):

$$p_{n+1} = \sum_i \binom{n}{i} p_i.$$

Show also that $\sum p_n x^n / n! = \exp[(\exp x) - 1]$.

(8) Define the natural ordering on $\mathbf{Z}$ in terms of the ordering of $\mathbf{N}$ and prove its compatibility with addition and multiplication.

(9) Show that there is no total ordering on $\mathbf{Z}/m$ preserving addition and multiplication.

(10) Use Th. 4 to give a proof that $\mathbf{N}$ is not finite.

(11) Give a direct proof by induction that there exists no surjective mapping from $\{1, 2, \dots, m\}$ to $\{1, 2, \dots, n\}$ if $m < n$.

1.2 Countable and uncountable sets

In 1.1 we gave a formal definition of finite sets; we now define a set to be *infinite* if it is not finite. Until relatively recent times the notion of 'infinity' was surrounded by a good deal of mystery and uncertainty, even in mathematics. Thus towards the middle of the last century, Bolzano propounded as a paradox the fact that (in modern terms) an infinite set might be equipotent to a proper subset of itself. A closer study reveals that every infinite set has this property, and this has even been taken as the basis of a definition of infinite sets; it certainly seems no longer a paradox. The work of Cantor, Dedekind and others from 1870 onwards has dispelled most of the uncertainties, and though mysteries remain they will not hamper us in the relatively straightforward use we shall make of the theory.

In order to extend the notion of counting to infinite sets, we associate with every set X , finite or not, an object $|X|$ called its *cardinal* or *cardinal number*, defined in such a way that two sets have the same cardinal iff they are equipotent. Such a definition is possible because, as we have seen, equipotence is an equivalence relation on any collection of sets.

A non-empty finite set has a natural number as its cardinal, the empty set has cardinal 0. All other sets are infinite, their cardinals are said to be *transfinite*. In particular, the set $\mathbf{N}$ of natural numbers is infinite; its cardinal is denoted by $\aleph_0$. The letter aleph: $\aleph$, the first of the Hebrew alphabet, is customarily used for cardinal numbers. A set of cardinal $\aleph_0$ is also said to be *countable* (or *enumerable*); thus A is countable iff there is bijection from $\mathbf{N}$ to A . If a set A is countable, we can write it in the form

$$A = \{a_1, a_2, a_3, \dots\}, \quad (1)$$

where the a_i are distinct. Such a representation of A is called an *enumeration* of A , and a proof that a set is countable will often consist in giving an

enumeration. Sometimes the term 'enumeration' is used for a set written as in (1) even if the a_i are not distinct; in that case we can always produce a strict enumeration by going through the sequence and omitting all repetitions. The set so obtained is finite or countable.

Many sets formed from countable sets are again countable, as our next result shows.

THEOREM 1 *Any subset and any quotient set of a countable set is finite or countable. Let A and B be countable sets, then $A \cup B$ and $A \times B$ are again countable; more generally the Cartesian product of any finite number of countable sets is countable. Further, the collection of all finite subsets of a countable set is countable.*

Proof. Any countable set A may be taken in the form (1); if A' is a subset, we go through the sequence $a_1, a_2, \dots$ and omit all terms not in A' to obtain an enumeration of A' . If A'' is a quotient, and $x \mapsto \bar{x}$ is the natural mapping from A to A'' , then

$$\{\bar{a}_1, \bar{a}_2, \dots\}$$

is an enumeration of A'' , possibly with repetitions; hence A'' is countable (or finite).

Next let A be given by (1) and let $B = \{b_1, b_2, \dots\}$, then $A \cup B$ may be enumerated as $\{a_1, b_1, a_2, b_2, a_3, \dots\}$, where repetitions (which may occur if $A \cap B \neq \emptyset$) may be discarded. To prove that $A \times B$ is countable, we arrange its elements as an (infinite) matrix:

$$\begin{array}{ccccccc} (a_1, b_1) & \rightarrow & (a_1, b_2) & \rightarrow & (a_1, b_3) & \rightarrow & \dots \\ & \swarrow & \downarrow & \searrow & \downarrow & \searrow & \\ (a_2, b_1) & & (a_2, b_2) & & (a_2, b_3) & & \dots \\ & \swarrow & \downarrow & \searrow & \downarrow & \searrow & \\ (a_3, b_1) & & (a_3, b_2) & & (a_3, b_3) & & \dots \\ \vdots & & \vdots & & \vdots & & \vdots \end{array}$$

which we enumerate by going along successive diagonals as indicated. Now the result for a product of r countable sets follows by induction on r .

Finally let A be any countable set and denote by A_r for $r = 1, 2, \dots$ the set of all r -element subsets of A . Clearly A_r is countable, for it may be mapped into the Cartesian power A^r by the rule

$$\{a_{i_1}, \dots, a_{i_r}\} \mapsto (a_{i_1}, \dots, a_{i_r}),$$

where $i_1, \dots, i_r$ is the sequence $i_1, \dots, i_r$ arranged in ascending order. This provides a bijection of A_r with a subset of A^r , and it follows that A_r is countable. Now $A_1 \cup A_2 \cup \dots$ is equipotent with $\mathbb{N}^2$, for if $A_r = \{c_{1r}, c_{2r}, \dots\}$, then $(i, j) \mapsto c_{ij}$ provides a bijection. Thus the collection of all non-empty finite subsets of A is countable, and adding $\emptyset$ as a further member we still have a countable set. ■

With the help of this result many sets can be proved to be countable which do not at first sight appear to be so. Thus $\mathbb{Z}$ can be written as a union of $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}' = \{0, -1, -2, \dots\}$; both $\mathbb{N}$ and $\mathbb{N}'$ are countable, hence so is $\mathbb{Z}$. The set $\mathbb{Q}_+$ of positive rational numbers is countable, as the image of $\mathbb{N}^2$ under the mapping $(a, b) \mapsto ab^{-1}$; now $\mathbb{Q}$ itself can be written as the union of the set of positive rational numbers, the negative rational numbers and 0; therefore $\mathbb{Q}$ is countable. The set of all algebraic numbers (cf. Ch. 6 below) is countable: for a given degree n , the set of all monic equations of degree n over $\mathbb{Q}$ is equipotent to $\mathbb{Q}^n$, if we map

$$(a_1, \dots, a_n) \mapsto x^n + a_1 x^{n-1} + \dots + a_n = 0.$$

Each equation has at most n complex roots, so the set S_n of all roots of equations of degree n is countable, and now the set of all algebraic numbers is $S_1 \cup S_2 \cup \dots$, which is again countable.

At this point a newcomer might be forgiven for thinking that perhaps every infinite set is countable. If that were so, there would of course be no need for an elaborate theory of cardinal numbers. In fact the existence of uncountable sets is one of the key results of Cantor's theory. Later (in Ex. (4)) we shall meet Cantor's diagonal argument which he used to prove the uncountability of the real numbers. The essential idea of this argument already appears in a more general result, also due to Cantor:

THEOREM 2 *Let X be any set and 2^X the set of all 2-valued functions on X , then 2^X is not equipotent with X .*

We shall prove that there is no surjective mapping from X to 2^X , it then follows that these sets are not bijective. Thus let $f: X \rightarrow 2^X$ be given, we must show that f is not surjective, i.e. we must find a function not in the image of f . In detail f associates with every $x \in X$ a 2-valued function on X , which may be denoted by f_x . We have to find a function $g: X \rightarrow 2$ which is different from all the f_x . This is easily done; taking the values as 0 and 1, we put

$$g(x) = \begin{cases} 1 & \text{if } f_x(x) = 0, \\ 0 & \text{if } f_x(x) = 1. \end{cases}$$

Then g is well-defined, and for each $x \in X$, $g \neq f_x$ because these functions differ at the point x . ■

If in this theorem we take X to be countable, then 2^X is again infinite, but uncountable. Moreover, we can in this way obtain arbitrarily large cardinals by starting from an infinite set X and forming in succession $2^X, 2^{2^X}, \dots$

Th. 2 again illustrates the dangers of operating with the 'set of all sets'. If we could form the union of all sets, U say, then U would contain 2^U as a subset, and that would enable us to map U onto 2^U (map each element of 2^U to itself, and any other element of U to some fixed member of 2^U). But this contradicts what has been proved in Th. 2. This paradox was discussed by

Burali-Fori and others in the closing years of the last century, and it provided the impetus for much of the axiomatic development that followed. Any axiom system now in use is designed to avoid the possibility of such paradoxes. For our purposes it is sufficient to note that we can avoid the paradoxes by not admitting constructions involving 'all sets' without further qualification.

Two particular applications of Th. 2 are worth recording. Let X be a set, then we denote by $\mathcal{P}(X)$ the set whose members are all the subsets of X ; e.g. $\mathcal{P}(\emptyset) = \{\emptyset\}$, $\mathcal{P}(\{x\}) = \{\emptyset, \{x\}\}$. This set $\mathcal{P}(X)$ is often called the *power set* of X ; it is equipotent with 2^X . To obtain a bijection we associate with each subset Y of X its *characteristic function*:

$$\chi_Y(x) = \begin{cases} 1 & \text{if } x \in Y, \\ 0 & \text{if } x \notin Y. \end{cases}$$

It is easily seen that this provides a bijection between $\mathcal{P}(X)$ and 2^X . The inverse is obtained by associating with each $f \in 2^X$ the inverse image of 1: $f^{-1}(1) = \{x \in X | f(x) = 1\}$. Now Th. 2 shows the truth of

COROLLARY 1 *Let X be any set, then its power set $\mathcal{P}(X)$ is not equipotent with X . More precisely, there is no surjection from X to $\mathcal{P}(X)$.* ■

COROLLARY 2 *The set $\mathbf{R}$ of real numbers is uncountable.*

Proof. It will be enough to find a subset of $\mathbf{R}$ which is equipotent to $2^{\aleph_0}$, since the latter is uncountable, the same will be true of $\mathbf{R}$. Consider those real numbers between 0 and 1 whose decimal expansions contain only 0s and 1s. If $a = 0 \cdot a_1 a_2 \dots$ is of this form, then f_a defined by $f_a(n) = a_n$ is a member of $2^{\aleph_0}$, and all members of $2^{\aleph_0}$ occur in this way. Thus we have found an uncountable set of real numbers. ■

A more precise analysis, given below, will show that $\mathbf{R}$ is actually equipotent with $2^{\aleph_0}$.

We have seen that cardinal numbers form an extension of the natural numbers. Now the set $\mathbf{N}$ has a structure, thus natural numbers can be compared, added and multiplied; it is reasonable to try to extend this structure to transfinite cardinals. If a, b are any cardinals, let A, B be sets such that $|A| = a$, $|B| = b$. We shall write $a \leq b$ whenever there is an injective mapping from A to B . Whether such a mapping exists depends clearly only on a, b and not on A, B themselves, so that the notation is justified. Further, $a \leq a$ holds for all a , and since the composition of two injections is an injection, it follows that $a \leq b$, $b \leq c$ imply $a \leq c$. Thus we have a preordering; this will in fact turn out to be a total ordering, but for the moment we content ourselves with proving that it is an ordering, i.e. that ' $\leq$ ' is antisymmetric. In terms of sets we must establish

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THEOREM 3 (Schröder-Bernstein theorem) *Let A, B be any sets and $f: A \rightarrow B$, $g: B \rightarrow A$ any injective mappings, then there is a bijection $h: A \rightarrow B$.*

Proof. Each element $a \in A$ is the image of at most one element of B under g , and each $b \in B$ is the image of at most one element of A under f . By alternating applications of f and g we produce an infinite sequence of successive images starting from $a \in A$: $a, af, afg, afgf, \dots$ and a sequence of inverse images which may or may not break off: $ag^{-1}, ag^{-1}f^{-1}, \dots$. If we trace a given element $a \in A$ as far back as possible we find one of three cases: (i) there is a first 'ancestor' in A , i.e. $a_0 \in A$, $a_0 \notin Bg$, such that $a = a_0(fg)^n$ for some n , (ii) there is a first ancestor in B , i.e. $b_0 \in B$, $b_0 \notin Af$, such that $a = b_0(gf)^n$ for some n , (iii) the sequence of inverse images continues indefinitely.

Each element of A comes under exactly one of these headings, and likewise each element of B . Thus A is partitioned into three subsets A_1, A_2, A_3 and similarly for B . Now it is clear that the restriction of f to A_1 is a bijection between A_1 and B_1 , for each element of B_1 comes from an element in A_1 . For the same reason the restriction of g to B_2 provides a bijection between B_2 and A_2 , and we can use either f (restricted to A_3) or g (restricted to B_3) to obtain a bijection between A_3 and B_3 . Thus we have found a bijection from A_i to B_i ($i = 1, 2, 3$) and putting these together we obtain a bijection between A and B . ■

This proof is essentially due to J. König (1906). As an application we determine the cardinal of the set $\mathbf{R}$ of real numbers. This cardinal is usually denoted by c and is called the *cardinal of the continuum*.

PROPOSITION 4 $c = 2^{\aleph_0}$.

Proof. The proof of Th. 2, Cor. 2 shows that $\mathbf{R}$ has a subset equipotent with $2^{\aleph_0}$, hence $2^{\aleph_0} \leq c$. On the other hand, $\mathbf{R}$ is equipotent with the real numbers in the open interval $(0, 1)$, e.g. $x \mapsto \frac{1}{2}(1+x^2)^{1/2} + x(1+x^2)^{-1/2}$ is a bijection. If we write each number as a fraction in the binary scale: $a = 0 \cdot a_1 a_2 \dots$, then $a \mapsto f_a$ where $f_a(n) = a_n$ is a mapping $(0, 1) \rightarrow 2^{\aleph_0}$ which is injective, for distinct real numbers have distinct binary fractions. Indeed, some have more than one, e.g. $0 \cdot 01001111 \dots$ and $0 \cdot 0101000 \dots$, but we can achieve uniqueness by excluding representations in which only finitely many digits are 0. It follows that $c \leq 2^{\aleph_0}$, and by Th. 3 the desired equality follows. ■

The sum and product of cardinals may be defined as follows. Let a, b be any cardinals, say $a = |A|$, $b = |B|$, and assume that $A \cap B = \emptyset$. Then $|A \cup B|$ depends only on a, b , not on A, B , and we define

$$a + b = |A \cup B|.$$

Similarly we put

$$ab = |A \times B|.$$

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It is easy to verify that these operations satisfy the commutative and associative laws, and a distributive law, as in the case of natural numbers. Moreover, for finite cardinals these operations agree with the usual definitions of addition and multiplication. On the other hand, the cancellation law does not hold, and there is nothing corresponding to subtraction or division. In fact, it can be shown that if $a, b \neq 0$ and at least one of a, b is infinite, then

$$a + b = ab = \max\{a, b\}. \quad (2)$$

Exercises

- (1) Show that the set of intervals in $\mathbf{R}$ with rational endpoints is countable.
- (2) Let A be an infinite set, A' a finite subset and B its complement in A . Show that $|A| = |B|$.
- (3) Let A be an uncountable set, A' a countable subset and B its complement in A . Show that $|A| = |B|$.
- (4) Fill in the details of the following alternative proof of Th. 2, Cor. 2: If the real numbers in binary form (as in the text proof of Cor. 2) could be enumerated $a^{(1)}, a^{(2)}, \dots$, we can find a number not included in the enumeration by putting $a = 0.b_1b_2\dots$ where $b_n = 0$ or 1 according as $a^{(n)}$ has 1 or 0 in the n th place. (This is Cantor's diagonal argument; b_n is chosen so as to differ from the diagonal term $a_n^{(n)}$).
- (5) Show that the set of all real functions on the interval $[0, 1]$ is greater than the cardinal of the continuum.
- (6) For any cardinals a, b, c , if $a \leq b$, show that $ac \leq bc$.
- (7) Define a^b as $|A^B|$, where $a = |A|$, $b = |B|$ and A^B is the set of all mappings from B to A . Check that this reduces to the definition of 2^b given in the text when $a = 2$, and prove that $a^b c^b = (abc)^b$, $a^b a^c = a^{b+c}$, $(a^b)^c = a^{bc}$.
- (8) Show that $a < 2^a$, for any cardinal a . More generally, show that $a < b^a$ if $b > 1$.

1.3 Zorn's lemma and well-ordered sets

We now take up the question of the comparability of cardinals left open in the last section, i.e. whether the ordering of cardinal numbers is in fact total. In terms of sets the question is whether, given two sets A, B , we can find an injective mapping from one to the other. In intuitive terms one might try to answer this by choosing an element from each of A and B , say a_1, b_1 , and pairing these off, then choosing another pair of elements a_2, b_2 and pairing them off, and so on. For sets that are at most countable this solves the problem, but we have seen that uncountable sets exist, and here the

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procedure adopted is rather more problematic. One way to overcome the difficulty is by introducing the concept of a well-ordering:

An ordered set A is said to be *well-ordered* if every non-empty subset of A has a least element.

A well-ordered set is always totally ordered, as we see by applying the definition to two-element subsets. A countable set may be well-ordered simply by enumerating its elements, e.g. the natural order of the positive integers is a well-ordering, but there are other well-orderings which do not put the countability into evidence, e.g. $\{2, 3, 4, \dots, 1\}$ or $\{1, 3, 5, \dots, 2, 4, 6, \dots\}$ or even $\{1, 2, 4, 6, \dots, 3, 9, 15, 21, \dots, 5, 25, 35, \dots, 7, 49, \dots\}$. Any well-ordering consists of a succession of sequences (possibly infinitely many). We have seen that a countable set can always be well-ordered, and soon we shall see what is needed to prove that every set can be well-ordered.

For well-ordered sets it is possible to prove the comparability in a strong form. Let us call a subset A' of an ordered set A a *lower segment* if $u \in A', v \leq u$ implies $v \in A'$. This definition can be used for any ordered set, not necessarily well-ordered (or even totally ordered). In particular, for any element $a \in A$, the set $L(a) = \{x \in A \mid x < a\}$ is a lower segment in A ; in a well-ordered set A every lower segment not the whole of A is of this form, for if A' is a proper lower segment of A and a is the first element of A not in A' , then $A' = L(a)$. Two ordered sets A, B are said to be *order-isomorphic* or of the same *order type* if there is a bijection between them which preserves the order, $f: A \rightarrow B$ such that $xf \leq yf \Leftrightarrow x \leq y$.

LEMMA 1 *A well-ordered set cannot be order-isomorphic to one of its proper lower segments.*

Proof. Let A be well-ordered, $L(a)$ a proper lower segment and let $f: A \rightarrow L(a)$ be an order-isomorphism. Then clearly $af < a$; let a_0 be the first element of A such that $a_0 f < a_0$. If we apply f to this inequality and recall that f preserves the order, we find that $a_0 f f < a_0 f$, so we have found an earlier element with the same property, namely $a_0 f$. This contradiction shows that f cannot exist. ■

We now show that any two well-ordered sets can be compared.

THEOREM 2 *Let A, B be two well-ordered sets, then one of them is order-isomorphic to a lower segment of the other.*

Proof. Let us call a pair of elements $a \in A$ and $b \in B$ *matched* if the corresponding lower segments $L(a)$ and $L(b)$ are order-isomorphic. Two distinct lower segments of A cannot be order-isomorphic, for one of them will be a lower segment of the other, and this would contradict Lemma 1. It follows that any element of B can be matched against at most one element of A and vice versa. Let A' be the set of elements of A that can be matched

against elements of B , and B' the set of elements of B matched against elements of A . Then A' and B' are order-isomorphic, as we see by using the correspondence provided by the matching. Moreover, A' is a lower segment of A , for if $a \in A'$ and $a_1 < a$, let a be matched to $b \in B$, then a_1 is matched to the element of $L(b)$ which corresponds to it under the isomorphism between $L(a)$ and $L(b)$. Similarly B' is a lower segment of B . If $A' \neq A$, then $A' = L(a')$ for some $a' \in A$; likewise if $B' \neq B$, then $B' = L(b')$ for some $b' \in B$, and by construction there is an order-isomorphism between $L(a')$ and $L(b')$, so that a' and b' are matched. But this is a contradiction (because $a' \notin L(a')$), therefore we have either $A' = A$ or $B' = B$ (or both) and the conclusion follows. ■

The problem of comparing cardinals is now reduced to the problem of well-ordering sets. If every set could be well-ordered, Th. 2 would tell us that any two cardinals are comparable. Now it was proved by Zermelo (in 1908) that every set can be well-ordered, but he had to make an assumption which was somewhat less intuitive than the other axioms used; this is the

AXIOM OF CHOICE *Given a family of non-empty sets $\{A_i\}_{i \in I}$ there exists a function which associates with each set A_i an element of A_i .*

At first sight this is an innocent-sounding assumption, which acquires its force from the fact that it applies to collections of sets with arbitrary indexing set. Only for finite families $\{A_1, \dots, A_n\}$ is no axiom needed. The axiom may be illustrated by the following example due to Bertrand Russell. A certain millionaire has infinitely many pairs of shoes and infinitely many pairs of socks. He wants to pick out one shoe from each pair: this causes no problems; he simply picks the left shoe each time. But when he wants to pick a sock from each pair, he needs the axiom of choice.

In some respects the axiom of choice occupies a position analogous to the parallel axiom in geometry (although set theory without the axiom of choice is not as interesting as non-Euclidean geometry). It has been proved consistent with and independent of the other axioms of set theory (by K. Gödel in 1939 and P. J. Cohen in 1963 respectively).

A logical step at this point would be to prove that every set can be well-ordered, using the axiom of choice. In fact we shall introduce another axiom, known as *Zorn's Lemma*, which is equivalent to the axiom of choice, and use it to prove the well-ordering theorem. This seems more appropriate in the present context, for it is Zorn's lemma rather than the axiom of choice that is used in algebra; we shall meet many examples later in this volume. By a *chain* in a partially ordered set we understand a totally ordered subset.

ZORN'S LEMMA *Let A be a non-empty partially ordered set; if every chain in A has an upper bound, then A has a maximal element.*

A partially ordered set is said to be *inductive* if it is non-empty and every chain in it has an upper bound; in this terminology Zorn's lemma states that every partially ordered set which is inductive has a maximal element.

This statement seems plausible, but any attempt at a direct proof soon encounters the situation typical of the axiom of choice. The actual proof, using the latter, can be found in most books on set theory. For an excellent account we refer to KAPLANSKY [1972]. Below, in Th. 3, we shall prove the well-ordering theorem (W) on the basis of Zorn's lemma (Z), and it is easy to prove the axiom of choice (C) from the well-ordering theorem: if $\{X_i\}$ is any family of non-empty sets, well-order $\bigcup X_i$ and assign to each X_i the element of it which is first in the given well-ordering. Thus the three assertions, W, Z, C are all equivalent.

Later on we shall meet many situations where the hypotheses of Zorn's lemma are satisfied; for the moment we shall give an illustration where the hypotheses do not hold. Let A be an infinite set and $\mathcal{F}$ the collection of all its finite subsets, partially ordered by inclusion. It is clear that $\mathcal{F}$ has no maximal element, and by Zorn's lemma this means that $\mathcal{F}$ contains chains which have no upper bound in $\mathcal{F}$; such chains are of course easily constructed. In verifying the hypotheses of Zorn's lemma it is important to test arbitrary chains and not merely ascending sequences, as is shown by examples (cf. Ex. (3)).

THEOREM 3 *Every set can be well-ordered.*

Proof. The idea of the proof is to consider well-orderings of parts of the given set, make these well-orderings into a partially ordered set and show that Zorn's lemma applies.

Thus let A be a set and $\mathcal{W}$ the collection of all well-ordered subsets of A . For any $X, Y \in \mathcal{W}$ we put $X \leq Y$ if X is a subset of Y and the inclusion mapping from X to Y is an order-isomorphism of X with a lower segment of Y ; in particular the ordering of X is then the same as that induced by Y . It is easily seen that $\mathcal{W}$ becomes an ordered set in this way. Let $\{X_\lambda\}$ be a chain in $\mathcal{W}$, where λ runs over an indexing set (not necessarily countable); thus for any λ, μ , either X_λ is a lower segment of X_μ or X_μ is a lower segment of X_λ . To obtain an upper bound for this chain we put $X = \bigcup X_\lambda$ and define an ordering on X as follows: let $x, y \in X$ and choose an index λ such that $x, y \in X_\lambda$. If $x \leq y$ in the ordering of X_λ , then the same is true in the ordering of X_μ for any μ such that $x, y \in X_\mu$, for of X_λ, X_μ one is a lower segment of the other, with the same ordering. Thus we may without ambiguity put $x \leq y$ in X if this holds in some X_λ , and the relation on X so defined is easily seen to be a well-ordering of X , with each X_λ as a lower segment. Hence X is an upper bound of the given chain.

We can now apply Zorn's lemma and obtain a maximal element X' in $\mathcal{W}$. We claim that $X' = A$; for if not, then there exists $z \in A, z \notin X'$. We form

$X'' = X' \cup \{z\}$ into a well-ordered set by taking the given order on X' and letting z follow all of X' . Then X'' is a member of $\mathcal{W}$ which is strictly greater than X' , a contradiction. Hence $X' = A$. ■

This result allows us to conclude that any two cardinals can be compared; thus the relation ' $\leq$ ' is a total ordering of cardinals. But we can say rather more than this. Theorems 2 and 3 suggest a classification of well-ordered sets according to their order type. Thus with every well-ordered set A we associate a symbol α , called its *ordinal number* or *order-type*, or simply *ordinal*, such that two well-ordered sets have the same ordinal precisely when they are order-isomorphic. Further we can define a relation $\alpha \leq \beta$ between ordinals, whenever a set of type α is order isomorphic to a lower segment of a set of type β . This is a well-defined relation on any set of ordinals, clearly transitive and by Lemma 1 antisymmetric, i.e. an ordering, which is total by Th. 2. In fact it is a well-ordering.

With each ordinal number α a cardinal number $|\alpha|$ may be associated, namely the cardinal of a well-ordered set of ordinal α . This is an order-preserving mapping from ordinals to cardinals, but not injective: to each finite cardinal there corresponds just one ordinal of the same type, but an infinite cardinal always corresponds to many different ordinals. However, we obtain a well-defined mapping by assigning to each cardinal the *least* ordinal which corresponds to it. E.g. a countable set, $\mathbb{N}$ say, may be well-ordered as $1, 2, 3, \dots$; this order-type is denoted by ω , it is the least countable ordinal. Another ordering, not isomorphic to the first, is $2, 3, 4, \dots, 1$; this is denoted by $\omega + 1$. Similarly, $n + 1, n + 2, \dots, 1, 2, \dots, n$ has ordinal $\omega + n$. The type of $1, 3, 5, \dots, 2, 4, 6, \dots$ is written $\omega + \omega$, and generally, given ordinals α, β we define $\alpha + \beta$ as the type of a well-ordered set of type α followed by one of type β (it is easily checked that such an arrangement gives rise to a well-ordered set whose type depends only on α and β). We observe that the addition of ordinal numbers is still associative, but not longer commutative: $1 + \omega = \omega \neq \omega + 1$.

We shall write 2ω for $\omega + \omega$ and generally $n\omega$ for $\omega + \dots + \omega$ to n terms. The limit of the sequence $\omega, 2\omega, 3\omega, \dots$ (i.e. the first ordinal following all of them) is written ω^2 ; we shall not pursue the topic further except to mention that every ordinal number α can be written in just one way as

$$\alpha = a_1\omega^{\alpha_1} + a_2\omega^{\alpha_2} + \dots + a_r\omega^{\alpha_r},$$

where $r, a_1, \dots, a_r$ are natural numbers and $\alpha_1, \dots, \alpha_r$ is a decreasing sequence of ordinal numbers (for a proof see, e.g., SIERPIŃSKI).

For well-ordered sets there is a form of induction known as the *principle of transfinite induction*. This is embodied in the remark (which practically reproduces the definition) that any non-empty subset of a well-ordered set has a least element. Let us pause briefly to examine how a transfinite induction proof looks in practice. One can distinguish three kinds of

ordinals: an ordinal number β may have an immediate predecessor α , then $\beta = \alpha + 1$, or it may have no immediate predecessor. In that case it is either the first ordinal 1, or it is the first ordinal after an infinite set of ordinals, in which case it is called a *limit ordinal*. Thus our three classes are (i) 1, (ii) ordinals which have an immediate predecessor, (iii) limit ordinals.

Now let A be a well-ordered set; if its ordinal is τ , the set is usually indexed by the ordinals less than τ : $A = \{a_\alpha\}_{\alpha < \tau}$. Suppose that a subset X satisfies the following conditions: (i) $a_1 \in X$, (ii) if $a_\alpha \in X$, then $a_{\alpha+1} \in X$ and (iii) if λ is a limit ordinal less than τ , and $a_\alpha \in X$ for all $\alpha < \lambda$, then $a_\lambda \in X$.

Then $X = A$. For if X were a proper subset of A , let a_β be the least element of A not in X , then $\beta > 1$, by (i). If β has an immediate predecessor α , then $a_\alpha \in X$ and $\beta = \alpha + 1$, hence by (ii), $a_\beta \in X$, a contradiction. So β must be a limit ordinal and by definition, $a_\alpha \in X$ for all $\alpha < \beta$. Hence by (iii) $a_\beta \in X$, again a contradiction; this proves that $X = A$, as asserted.

We conclude this section with the proof of a special case of a formula mentioned earlier ((2), 1.2).

PROPOSITION 4 For any infinite cardinal a ,

$$\aleph_0 a = a. \quad (1)$$

Proof. We shall need the associative law of multiplication of cardinal numbers: $(ab)c = a(bc)$. This is easily proved if we observe that each side may be regarded as the cardinal number of the product $A \times B \times C$, where A, B, C are sets of cardinals a, b, c respectively.

In the case where $a = \aleph_0$, (1) states that $\aleph_0^2$ is equipotent with $\aleph_0$, and this was proved in Th. 1 of 1.2. Secondly, if a is of the form $\aleph_0 b$, then

$$\aleph_0 a = \aleph_0(\aleph_0 b) = \aleph_0^2 b = \aleph_0 b = a,$$

which proves (1) in this case. We complete the proof by showing that every infinite cardinal is of the form $\aleph_0 b$. This amounts to showing that every infinite set A is equipotent with a set of the form $\mathbb{N} \times B$, for a suitable set B .

Let A be an infinite set, then by Th. 3, A can be well-ordered. Arranged in this form A consists of a totally ordered (in fact, well-ordered) set of countable sequences, followed by a finite (possibly empty) sequence. Since A is infinite, at least one infinite sequence occurs, and we may rearrange A by taking the finite sequence from the end and putting it in front of the first sequence. The set A now consists entirely of countable sequences, i.e. well-ordered sets of type ω . If they are indexed by a set B , it follows that A is equipotent with $\mathbb{N} \times B$, as we wished to show. ■

Exercises

- (1) Show that the axiom of choice is equivalent to the following axiom: Every surjective mapping has a left inverse.

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- (2) Let Φ be a partial ordering relation on a set A . Show that there is a total order Φ' on A such that $\Phi \subseteq \Phi'$.
- (3) Let A be an uncountable set and $\mathcal{F}$ the collection of all its countable subsets. Show that every ascending (countable) sequence of members of $\mathcal{F}$ has an upper bound in $\mathcal{F}$, but that $\mathcal{F}$ has no maximal element.
- (4) Show that any totally ordered set X has a well-ordered subset Y with the property: for each $x \in X$ there exists $y \in Y$ such that $x \leq y$.
- (5) Determine all order-automorphisms of $\mathbf{Z}$ (i.e. all order-preserving permutations of $\mathbf{Z}$).
- (6) Find a well-ordering of the set $\mathbf{Z}$ of all integers. Find well-orderings of $\mathbf{N}$ of type 2ω , $2\omega + 1$, $\omega^2 + 1$.
- (7) Show that any set of ordinals is well-ordered by $\leq$.
- (8) Check that $\alpha + \beta$ is well-defined and satisfies the associative law.
- (9) Ordinal multiplication may be defined by taking, for any ordinals α, β , sets A, B of type α, β respectively, and denoting by $\alpha\beta$ the ordinal of $A \times B$, ordered lexicographically. Show that this multiplication is well-defined, and that it agrees with the following definition: (i) $1\beta = \beta$, (ii) $(\alpha + 1)\beta = \alpha\beta + \beta$, (iii) if λ is a limit ordinal, then $\lambda\beta = \sup\{\gamma\beta \mid \gamma < \lambda\}$.
- (10) Show that $(\alpha + \beta)\gamma = \alpha\gamma + \beta\gamma$, $\alpha(\beta\gamma) = (\alpha\beta)\gamma$, but that in general, $\alpha(\beta + \gamma) \neq \alpha\beta + \alpha\gamma$. (Hint. For the inequality take β, γ finite and α infinite.)
- (11) From the inductive definition show that if $\alpha < \beta$, $\gamma > 0$, then $\alpha\gamma < \beta\gamma$. Deduce that $\gamma \neq 0$ and $\alpha\gamma = \beta\gamma$ implies $\alpha = \beta$.
- (12) Show that if β is a limit ordinal, then so is $\alpha + \beta$, for any ordinal α . Is $\beta + \alpha$ necessarily a limit ordinal? If $\alpha > 0$ and β is a limit ordinal, show that $\alpha\beta$ is a limit ordinal.

1.4 Abstract dependence relations

The notion of linear dependence in a vector space has certain properties which we shall meet again later when we come to discuss algebraic dependence (Ch. 6). These properties also occur elsewhere, e.g. in some geometrical constructions, and it is convenient to single them out here and discuss their properties in general. At the same time we shall provide a proof, as promised in Vol. 1, that every vector space, even infinite-dimensional, has a basis.

Let S be a set; by a *dependence relation* on S we understand a rule which associates with each finite subset X of S certain elements of S , said to be *dependent* on X . Further, the following conditions are satisfied:

D. 0 If $X = \{x_1, \dots, x_n\}$, then each x_i is *dependent* on X .

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- D. 1 (Transitivity):** If z is dependent on $\{y_1, \dots, y_m\}$ and each y_i is dependent on $\{x_1, \dots, x_n\}$, then z is dependent on $\{x_1, \dots, x_n\}$.
- D. 2 (Exchange property):** If y is dependent on $\{x_1, \dots, x_n\}$, but not on $\{x_2, \dots, x_n\}$, then x_1 is dependent on $\{y, x_2, \dots, x_n\}$.

E.g. the notion of linear dependence over a field satisfies **D. 0-2**, as is easily verified. In a module over a general ring R , the notion of being a linear combination of the elements of X is a relation satisfying **D. 0, 1**, but not in general **D. 2**. Similarly the notion of 'satisfying a non-trivial linear relation' with the elements of X is a relation satisfying **D. 0, 2** but not in general **D. 1**. Of course when R is a field (even skew) these two notions coincide with linear dependence and satisfy **D. 0-2**.

We note by **D. 0, 1**, if y is dependent on X' and $X' \subseteq X$, then y is dependent on X .

We shall extend the general notion of dependence to infinite subsets by saying that y is dependent on an infinite set X if there is a finite subset X' of X such that y is dependent on X' . Then it is easily seen that the exchange property **D. 2** holds for arbitrary sets X , not necessarily finite.

Given a dependence relation on a set S , we define for each $X \subseteq S$, the *span* of X as the set $\langle X \rangle$ of all elements of S dependent on X ; we also say that $\langle X \rangle$ is *spanned* by X . A family $X = \{x_i \mid i \in I\}$ of elements of S is said to be *independent* if no x_i is dependent on the remaining members of X ; otherwise X is called *dependent*. Clearly an independent family consists of distinct elements. An independent family which spans S is called a *basis* of S . It is clear that this generalizes the definitions given in Vol. 1 for vector spaces and, as in that case, we can recognize bases by the following result:

PROPOSITION 1 Let S be a set with a dependence relation, then for any subset X of S the following three conditions are equivalent:

- (a) X is a maximal independent subset of S ,
- (b) X is a minimal spanning set of S ,
- (c) X is a basis of S .

Proof. (a) $\Leftrightarrow$ (c). Let X be a maximal independent subset of S , then any $x \in X$ is dependent on X by **D. 0**, and if $y \notin X$, then $X \cup \{y\}$ is dependent, by maximality, so some element is dependent on the rest, say $x \in X$ is dependent on $X' \cup \{y\}$, where X' is the complement of $\{x\}$ in X . Since X is independent, x is not dependent on X' , hence by **D. 2**, y is dependent on $X = X' \cup \{x\}$, and this shows that X spans S . Conversely, if X is a basis, it is independent, but every element not in X is dependent on it, so X is maximal independent.

(b) $\Leftrightarrow$ (c). Let X be a minimal spanning set; if it were dependent, say $x \in X$ is dependent on the rest of X , then we could omit x and still have a spanning

set, by **D. 1**; this contradicts minimality, hence X is a basis. Conversely, if X is a basis, it is independent, so no element is dependent on the rest, and X is a minimal spanning set. ■

It is instructive to go through this proof in a particular case, and the reader is advised to do this, say for vector spaces over a skew field, taking first the case where the basis is finite and then the general case.

Our main aim will be to show that every set with a dependence relation has a basis, and that any two bases have the same number of elements. This will also show that every vector space, even infinite-dimensional, has a basis, and a uniquely defined dimension. First we need a replacement for the exchange lemma (p. 73, Vol. 1).

LEMMA 2 (Exchange lemma) *Let S be a set with a dependence relation. If X is an independent subset and Y a spanning set of S , then we can complete X to a basis of S by elements from Y , i.e. there is a subset Y' of Y such that $X \cup Y' = \emptyset$ and $X \cup Y'$ is a basis of S . Moreover, if Y is finite, then $|X \cup Y'| \leq |Y|$.*

Proof. Consider the collection of all independent subsets Z of S such that

$$X \subseteq Z \subseteq X \cup Y. \quad (1)$$

Since X is independent, there are such sets Z . Moreover, if $\{Z_\alpha\}$ is any chain of such subsets, then the union $Z = \bigcup Z_\alpha$ again satisfies (1), and it is independent, for if not, then some $z \in Z$ would be dependent on a finite subset of $Z \setminus \{z\}$, say on $\{z_1, \dots, z_r\}$, but $z, z_1, \dots, z_r$ all lie in some Z_α and this contradicts the independence of Z_α . By Zorn's lemma the collection has a maximal member B , i.e. a maximal independent set B such that $X \subseteq B \subseteq X \cup Y$. This shows that B consists of X together with certain elements of Y . By maximality, every element of Y depends on B , hence $\langle B \rangle = \langle Y \rangle = S$, therefore B is a basis of S , as claimed.

Now assume that Y is finite, then any maximal independent subset of Y spans S , hence is a basis. We shall show that starting from any finite subset X' of X we can obtain a basis by adjoining at most $|Y| - |X'|$ elements of Y . This will show that $|X| \leq |Y|$ and when $X' = X$ we reach the desired conclusion $|X \cup Y'| \leq |Y|$. We shall use induction on $|X'|$, thus suppose we have a basis $B_{r-1} = \{x_1, \dots, x_{r-1}, y_1, \dots, y_n\}$, where $x_i \in X, y_j \in Y$. If X contains another element x_r , then $B_{r-1} \cup \{x_r\}$ is a spanning set containing the independent subset $\{x_1, \dots, x_r\}$. By the first part it contains a basis B_r including $x_1, \dots, x_r$ but B_r cannot contain all of $y_1, \dots, y_n$ because the family $\{x_1, \dots, x_r, y_1, \dots, y_n\}$ is dependent (x_r is dependent on the rest). Hence B_r contains at most n elements, i.e. $|B_r| \leq |B_{r-1}|$. Now the conclusion follows by induction on r . ■

By the last part, the number of elements in an independent family is bounded by the number of elements in a finite spanning set. In particular,

given two bases B, C , if B is finite, then $|C| \leq |B|$ and by symmetry $|B| \leq |C|$. Thus we have the

COROLLARY *Let S be a set with a dependence relation. If S has a finite spanning set C , then any independent family has at most $|C|$ members and any two bases have the same number of elements.* ■

Our next task is to show that any two bases, finite or not, have the same number of elements. For the case of infinite bases this is true in a rather wider context, which we shall explain briefly, as it is sometimes useful.

Suppose that instead of a dependence relation we have a relation satisfying only **D. 0** and **D. 1** (not necessarily **D. 2**); let us call this a *spanning relation*. We can as before define the *span* of a subset X as the set $\langle X \rangle$ of elements dependent on X . If we examine the proof of Prop. 1 we find that for a spanning relation (c) $\Leftrightarrow$ (b) $\Rightarrow$ (a), but a maximal independent set need no longer be a basis. In fact the term 'basis' in this context tends to mislead and we shall only speak of minimal spanning sets. The example of the rational numbers $\mathbf{Q}$ (as $\mathbf{Z}$ -module) shows that minimal spanning sets need not exist. But when they do, we have the following relation:

PROPOSITION 3 *Let S be a set with a spanning relation, and let X be a minimal spanning set of S . If X is infinite, of cardinal a , then any spanning set of S has at least a elements. In particular, S has no finite spanning set, and any two minimal spanning sets of S have the same cardinal.*

Proof. Let Y be any spanning set of S and denote its cardinal by b . Each $y \in Y$ is dependent on X and hence is dependent on a finite subset X_y of X . We claim that

$$X = \bigcup_{y \in Y} X_y. \quad (2)$$

For clearly $\bigcup X_y \subseteq X$, and since $y \in \langle X_y \rangle$, it follows that $\langle \bigcup X_y \rangle \supseteq \langle Y \rangle = S$. Thus $\bigcup X_y$ is a subset of X spanning S and (2) follows by the minimality of X . If Y were finite, (2) would express X as the union of a finite number of finite sets, which contradicts the fact that X is infinite. Hence Y is infinite, and by (2) and Prop. 4 of 1.3, we find

$$a = |X| \leq \sum |X_y| \leq \aleph_0 b = b.$$

This shows that Y has cardinal at least a . If Y is also minimal, then by interchanging the roles of X and Y we see that $b \leq a$, and hence $b = a$. ■

As we observed earlier, the relation ' y is in the submodule generated by X ' is a spanning relation; we therefore have the

COROLLARY *Let R be a ring and M any R -module. If M has a minimal generating set X which is infinite, then every generating set of M is infinite, of*

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cardinal at least $|X|$, and all minimal generating sets of M have the same cardinal. ■

Finally we have

THEOREM 4 Let S be a set with a dependence relation, then S has a basis. More precisely, if X is an independent subset and Y a spanning set of S such that $X \subseteq Y$, then there is a basis B of S such that

$$X \subseteq B \subseteq Y, \quad (3)$$

and any two bases have the same cardinal.

Proof. Let X, Y be as stated, then $X \cup Y = Y$ and by Lemma 2 there exists a basis satisfying (3). In particular, taking $X = \emptyset$, $Y = S$ we see that there always is a basis. The last part follows by Lemma 2, Cor., and Prop. 3. ■

As a first application we have

COROLLARY 1 Any vector space over a field (even skew) has a basis, and any two bases have the same cardinal. ■

E.g. $\mathbf{R}$ considered as a $\mathbf{Q}$ -space has a basis, but the proof (involving Zorn's lemma) is non-constructive, and no explicit instance of such a basis is known. Any $\mathbf{Q}$ -basis of $\mathbf{R}$ is called a *Hamel basis* of the real numbers.

Later we shall on one occasion (in Ch. 12) have to deal with a relation satisfying D. 0, 2 but only a weakened form of D. 1 obtained by assuming $\{y_1, \dots, y_m\}$ independent, thus our relation satisfies

D. 1' If z is dependent on the independent set $\{y_1, \dots, y_m\}$ and each y_i is dependent on X , then z is dependent on X .

We still find that bases are just maximal independent sets, for the proof (a) $\Leftrightarrow$ (c) of Prop. 1 did not use D. 1, and maximal independent sets always exist, by Zorn's lemma, so every independent subset is contained in a basis. Now the proof of Lemma 2, in the case where X, Y are finite bases, uses D. 1 only in the weaker form D. 1', so we still have the following conclusion:

COROLLARY 2 Let S be a set with a relation satisfying D. 0, 1', 2 then any independent subset is contained in a basis, and if S has a finite basis, any two bases have the same cardinal. ■

The axiomatic study of dependence relations turns out to have connexions with many other fields, in particular graph theory, combinatorial theory, block designs, etc. See D. J. A. WELSH.

Exercises

- (1) Prove the exchange property for infinite sets: If y is dependent on $X \cup \{z\}$ but not on X , then z is dependent on $X \cup \{y\}$.

Further exercises

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- (2) Check that in the proof of Prop. 1, (c) $\Rightarrow$ (a), (b) only uses D. 0, (a) $\Rightarrow$ (c) only uses D. 0, 2 and (b) $\Rightarrow$ (c) only uses D. 0, 1.
- (3) Let S be a set with a dependence relation and for each subset X write $r(X)$ for the cardinal of a basis of X . Show that (i) $r(\emptyset) = 0$, (ii) $r(X) \leq r(X \cup \{y\}) \leq r(X) + 1$, (iii) if $r(X) = r(X \cup \{y\}) = r(X \cup \{z\})$, then $r(X \cup \{y, z\}) = r(X)$. Show that any function $r(X)$ on the subsets of a set S with integer values, satisfying (i)–(iii) leads to a dependence relation on S .
- (4) Show that for any $n \geq 1$ there is a minimal spanning set for $\mathbf{Z}$ (as $\mathbf{Z}$ -module) of n elements.
- (5) How many dependence relations are there on a set of three elements?
- (6) Let S be a set with a dependence relation. A subset of S is said to be *closed* if it contains all the elements dependent on it. Show that the intersection of any family of closed sets is closed. Find all dependence relations for which the union of any two closed sets is closed.
- (7) Let S be a set with a spanning relation (D. 0, 1) and let A be the set of all $a \in S$ which can be omitted from any spanning set, i.e. if $X \cup \{a\}$ spans S , then so does X . Show that A is the intersection of all maximal closed proper subsets of S .
- (8) Let S be a set with a dependence relation. Show that the relation is determined by the collection of bases of S .
- (9) (Whitney) Let S be a set with a dependence relation Φ . Let Φ^* be the set of complements of bases of S for the relation Φ . Show that there is a dependence relation on S whose collection of bases is Φ^* . (Hint. Show that there is a dependence relation whose independent sets are the complements of the spanning sets of S .)
- (10) Show that every additive mapping of $\mathbf{R}$ into itself is $\mathbf{Q}$ -linear, but not necessarily $\mathbf{R}$ -linear. (Hint. Use a Hamel basis.)

Further exercises on Chapter 1

- (1) For any integers $r \geq 0$, $m > 0$, define an induction algebra $N(r, m)$ on the set $\{1, 2, \dots, r+m\}$ by the rule

$$i' = \begin{cases} i+1 & \text{if } i < r+m, \\ i+1-m & \text{if } i = r+m. \end{cases}$$

Show that $N(r, m)$ satisfies N. 1, 2, 5 for all r, m , and that N. 3 holds iff $r > 0$, while N. 4 holds for no values of r, m .

- (2) Show that every induction algebra on a single generator is either isomorphic to $\mathbf{N}$, or to one of the algebras $N(r, m)$ defined in Ex. (1).

- (3) Prove the division algorithm from the Peano axioms: Given $a, b \in \mathbf{N}$ there exist q, r such that $a = bq + r$, where $0 \leq r < b$.

- (4) Let A and B be finite sets of a, b elements respectively. Show that the number $f(a, b)$ of mappings of B into A can be defined recursively by the equations $f(a, 1) = a$, $f(a, b) = f(a, b-1)f(a, b)$. Prove the relations $f(a, b+c) = f(a, b)f(a, c)$, $f(ab, c) = f(a, c)f(b, c)$. Give a definition of $f(a, b)$ by recursion in the first variable. (Hint. Use the binomial theorem.)
- (5) Show that the union of a finite family of finite sets is finite.
- (6) If a, b are cardinals of which at least one is infinite, show that $a + b = \max\{a, b\}$.
- (7) Show that a set is finite iff every collection of subsets has either a maximal or a minimal element.
- (8) (Sierpinski) A collection T of sets will (in this exercise) be called a *tower* if $\emptyset \in T$ and $X, Y \in T \Rightarrow X \cup Y \in T$. Show that a set A is finite iff A belongs to every tower T such that $\{x\} \in T$ for all $x \in A$.
- (9) Prove that Zorn's lemma is equivalent to Hausdorff's maximal principle: Given a collection $\mathcal{A}$ of sets, any chain $\mathcal{C}$ in $\mathcal{A}$ is contained in a maximal chain.
- (10) (Tarski) Show that a set is finite iff it can be ordered so that both the ordering and its opposite are well-orderings.
- (11) Define an *upper segment* of an ordered set as a lower segment in the opposite ordering. Verify that the upper segments are precisely the complements of the lower segments. If a set A is well-ordered, and order-isomorphic to every non-empty upper segment of itself, what can be said about A ? Show that A cannot be of the type 2ω .
- (12) Let Γ be a graph (i.e. a finite set of points, and certain pairs of points are joined by lines). Show that there exists a dependence relation on Γ in which the minimal dependent sets are the points occurring in some closed circuit.
- (13) Verify that the relation of being algebraically dependent over a field is a dependence relation (cf. Ch. 6). Let $S = \{x, y, xy, x+y\}$ be the subset of $\mathbb{F}_2(k, y)$, where x, y are indeterminates over $\mathbb{F}_2$. Show that under algebraic dependence the independent sets are the subsets of S of at most two elements. Show that there is no vector space over $\mathbb{F}_2$ with a subset $\{a, b, c, d\}$ such that the linearly independent subsets are precisely the subsets of cardinal ≤ 2 .
- (14) (Mac Lane–Ingelton) Let $S = \{1, \dots, 9\}$ and consider the dependence relation with minimal dependent families $\{1, 2, 3\}$, $\{4, 5, 6\}$, $\{1, 7, 5\}$, $\{1, 8, 6\}$, $\{2, 7, 4\}$, $\{2, 9, 6\}$, $\{3, 8, 4\}$, $\{3, 9, 5\}$. Show that there is no vector space with this dependence relation. (Hint. Use Pappus' theorem to show that $\{7, 8, 9\}$ would have to be dependent.)

2

Lattices

2.1 Definitions; modular and distributive lattices

In any partially ordered set S an element c is a *greatest* element if $x \leq c$ for all $x \in S$, while c is called *maximal* if $c < x$ for no $x \in S$. Thus a greatest element is maximal but the converse need not hold; a greatest element, if it exists, is clearly unique (unlike a maximal element). *Least* and *minimal* elements are defined dually. E.g. $\mathbb{N}$ has a least element, but no greatest element while $\mathbb{Q}$ has neither a least nor a greatest element.

An *upper bound* of a subset X of S is an element $b \in S$ such that $x \leq b$ for all $x \in X$; here b may or may not belong to X . *Lower bounds* are defined dually and a set is *bounded* if it has both an upper and a lower bound.

Both $\mathbb{N}$ and $\mathbb{Q}$ are totally ordered, i.e. $a \leq b$ or $b \leq a$ for all a, b ; most partially ordered sets do not share this property, e.g. the set $\mathcal{P}(S)$ of all subsets of a set S (with more than one element) relative to inclusion, or $\mathbb{N}$ relative to divisibility. In both these cases any two elements have a *least upper bound* or *supremum*, briefly sup, i.e. given x, y , there exists z such that (i) $x \leq z$, $y \leq z$ and (ii) $x \leq z'$, $y \leq z'$ implies $z \leq z'$. Dually, any two elements have a *greatest lower bound* or *infimum*, briefly inf, defined similarly. Thus if X and Y are subsets of S , their sup is $X \cup Y$ and their inf is $X \cap Y$; if m, n are natural numbers, their sup (relative to divisibility) is their least common multiple while their inf is their highest common factor.

A partially ordered set in which any two elements have a supremum and an infimum is called a *lattice*. The sup of x, y is written $x \vee y$ and is called the *join* of x and y ; the inf of x, y is written $x \wedge y$ and is called the *meet* of x and y . Clearly every totally ordered set is a lattice; so is the set $\mathbb{N}$ of natural numbers under divisibility, as well as the set $\mathcal{P}(S)$ of subsets of S , under inclusion.

A lattice may be regarded as a set with two binary operators, $\vee$ and $\wedge$. These operators satisfy a number of laws, reminiscent of the laws of addition and multiplication, and these laws can be used to give an alternative definition of lattices.

PROPOSITION 1 Let L be a lattice, then for any $a, b, c \in L$,
 $a \vee (b \vee c) = (a \vee b) \vee c$, $a \wedge (b \wedge c) = (a \wedge b) \wedge c$ (associative law), (1)